

# Supplement

## Bayesian Adaptive Deep Reinforcement Learning for Personalized Treatment in Multiple Myeloma

Orlane Rossini<sup>1</sup>, Meritxell Vinyals<sup>2</sup>, Alice Cleynen<sup>1,3</sup>, Benoîte de Saporta<sup>1</sup>, and  
Régis Sabbadin<sup>2</sup>

<sup>1</sup> IMAG, Univ Montpellier, CNRS, Montpellier, France

<sup>2</sup> Univ Toulouse, INRAE-MIAT, Toulouse, France

<sup>3</sup> IRL FAMSI, Australian National University, Canberra, ACT, Australia

### 1 PDMP framework details

Let us first describe the *controlled PDMP* model  $\langle E, (\Phi, \lambda, Q)^\ell \rangle$  for the medical scenario.

The **state space**  $E = \{E_m\}_{m \in M}$  is defined as follows, with  $M = \{0, 1, 2, 3\} \times \{0, 1, \dots, K\}$  representing the mode part. A mode  $m = (h, k) \in M$  is composed of a *health status*,  $h$  ( $h = 0$ : remission,  $h = 1$ : curable relapse,  $h = 2$ : non curable relapse,  $h = 3$ : death) and of a number,  $k \in \{0, \dots, K\}$ , of past applied treatments.

Subsets  $E_m$  are defined as follows:

- $E_{m=(0,k)} = \{\zeta_0\} \times [0, T]^2$ ,  $\forall k$ : a patient in remission (health status  $h = 0$ ) has a marker level of  $\zeta_0$ , whatever his current treatment and number of past treatments.
- $E_{m=(1,k)} = [\zeta_0, D) \times [0, T]^2$ ,  $\forall k$ : a patient suffering a relapse (neither dead nor in remission) has a marker level  $\zeta_0 \leq \zeta < D$ , whatever the number of past treatments.
- $E_{m=(2,k)} = [\zeta_0, D) \times [0, T]^2$ ,  $\forall k$ : a patient suffering an incurable relapse (neither dead nor in remission) has a marker level  $\zeta_0 \leq \zeta < D$ , whatever the number of past treatments.

Let  $\{(\Phi, \lambda, Q)^\ell : \ell \in \{0, 1\}\}$  denote the finite collection of available dynamics for the PDMP on the state space  $E$ . The case  $\ell = 0$  corresponds to the dynamics without treatment, while  $\ell = 1$  corresponds to the dynamics under treatment.

**Dynamics without treatment** ( $\ell = 0$ ). The biomarker evolution follows an exponential pattern. Under remission ( $h = 0$ ), it remains constant at  $\zeta_0$ , while for a relapse ( $h = 1$  and  $h = 2$ ), it is increasing:

$$\begin{aligned}\Phi_{(0,k)}^0(x = (\zeta, u, \tau), s) &= (\zeta_0, u + s, 0), \\ \Phi_{(1,k)}^0(x = (\zeta, u, \tau), s) &= (\zeta e^{v_1 s}, u + s, 0), \\ \Phi_{(2,k)}^0(x = (\zeta, u, \tau), s) &= (\zeta e^{v_2 s}, u + s, 0).\end{aligned}$$

The time required for the process to reach its boundary is infinite under remission, and, under relapses, correspond to the time required to reach the death biomarker level  $D$ :

$$\begin{aligned} t_{(0,k)}^{0,*}(\mathbf{x}) &= +\infty, \\ t_{(1,k)}^{0,*}(\mathbf{x}) &= \frac{1}{v_1} \log\left(\frac{D}{\zeta}\right), \\ t_{(2,k)}^{0,*}(\mathbf{x}) &= \frac{1}{v_2} \log\left(\frac{D}{\zeta}\right). \end{aligned}$$

From remission, the probability of relapse occurrence increases with the duration of time spent in remission and the number of previous treatments. From standard relapse, the risk of relapses associated with therapeutic escape is influenced by the biomarker level and the number of previous treatments. In light of these considerations, we choose Weibull distributions of the form:  $\mu_{\mathbf{m} \rightarrow (h',k)}^0(u) = (\alpha_{\mathbf{m} \rightarrow (h',k)}^0 u)^{\beta_{\mathbf{m} \rightarrow (h',k)}^0}$  and  $\mu_{\mathbf{m} \rightarrow (2,k)}^0(\zeta) = (\alpha_k^0 \zeta)^{\beta_0^0}$ :

$$\begin{aligned} \lambda_{(0,k)}^0(\mathbf{x}) &= (\mu_{\mathbf{m} \rightarrow (1,k)}^0 + \mu_{\mathbf{m} \rightarrow (2,k)}^0)(u), \\ \lambda_{(1,k)}^0(\mathbf{x}) &= \mu_{\mathbf{m} \rightarrow (2,k)}^0(\zeta), \\ \lambda_{(2,k)}^0(\mathbf{x}) &= 0. \end{aligned}$$

In remission, the patient may transition to either a standard relapse or an incurable relapse. In the case of relapse or therapeutic escape, the biomarker increases to a critical value  $D$ , leading to the patient's death. A therapeutic escape may occur at any time. We define the Markov kernel  $Q$  (case  $h = 3$  is omitted as no jumps are allowed when patients are dead) for all real-valued bounded measurable functions  $g$  on  $E$ :

$$\begin{aligned} Q_{(0,k)}^0 g(x) &= g(1, k, \zeta_0, 0, 0) \frac{\mu_{\mathbf{m} \rightarrow (1,k)}^0(u)}{(\mu_{\mathbf{m} \rightarrow (1,k)}^0 + \mu_{\mathbf{m} \rightarrow (2,k)}^0)(u)} \\ &\quad + g(2, k, \zeta_0, 0, 0) \frac{\mu_{\mathbf{m} \rightarrow (2,k)}^0(u)}{(\mu_{\mathbf{m} \rightarrow (1,k)}^0 + \mu_{\mathbf{m} \rightarrow (2,k)}^0)(u)}, \\ Q_{(1,k)}^0 g(x) &= g(2, k, \zeta, 0, 0) \mathbb{1}_{D > \zeta} + g(3, k, D, 0, 0) \mathbb{1}_{\zeta = D}, \\ Q_{(2,k)}^0 g(x) &= g(3, k, D, 0, 0) \mathbb{1}_{\zeta = D}. \end{aligned}$$

**Dynamics with treatment** ( $\ell = 1$ ). The biomarker evolution follows an exponential pattern. Under remission ( $h = 0$ ), it remains constant at  $\zeta_0$ , while for a relapse  $h = 1$  it is decreasing, and for incurable relapse ( $h = 2$ ), it is increasing:

$$\begin{aligned} \Phi_{(0,k)}^1(\mathbf{x} = (\zeta, u, \tau), s) &= (\zeta_0, u + s, \tau + s), \\ \Phi_{(1,k)}^1(\mathbf{x} = (\zeta, u, \tau), s) &= (\zeta e^{-\frac{v_1}{k}s}, u + s, \tau + s), \\ \Phi_{(2,k)}^1(\mathbf{x} = (\zeta, u, \tau), s) &= (\zeta e^{v_2 s}, u + s, \tau + s). \end{aligned}$$

The time required for the process to reach its boundary is infinite under remission, and, under relapse, corresponds to the time required to reach the nominal value  $\zeta_0$  while under incurable relapse it is the time required to reach the death biomarker level  $D$ :

$$\begin{aligned} t_{(0,k)}^{1,*}(\mathbf{x}) &= +\infty, \\ t_{(1,k)}^{1,*}(\mathbf{x}) &= \frac{k}{v_1} \log\left(\frac{\zeta}{\zeta_0}\right), \\ t_{(2,k)}^{1,*}(\mathbf{x}) &= \frac{1}{v_2} \log\left(\frac{D}{\zeta}\right). \end{aligned}$$

From remission, the probability of incurable relapse occurrence increases with the duration of time spent in remission and the number of previous treatments. From standard relapse, the risk of relapses associated with therapeutic escape is influenced by the biomarker level and the number of previous treatments. In light of these considerations, we, once again, choose Weibull distributions of the form:

$$\mu_{(0,k) \rightarrow (h',k)}^1(u) = (\alpha_{(0,k) \rightarrow (h',k)}^1 u)^{\beta_{(0,k) \rightarrow (h',k)}^1} \text{ and } \mu_{(1,k) \rightarrow (2,k)}^1(\zeta) = (\alpha_k^1 \zeta)^{\beta^1}:$$

$$\begin{aligned} \lambda_{(0,k)}^1(\mathbf{x}) &= \mu_{(0,k) \rightarrow (2,k)}^1(u), \\ \lambda_{(1,k)}^1(\mathbf{x}) &= \mu_{(1,k) \rightarrow (2,k)}^1(\zeta), \\ \lambda_{(2,k)}^1(\mathbf{x}) &= 0. \end{aligned}$$

In remission, the patient may transition to an incurable relapse. In the case of relapse the biomarker decreases to  $\zeta_0$  and returns to remission. Therapeutic escape may occur at any time. In the case of therapeutic escape, the biomarker increases toward the  $D$  threshold, ultimately resulting in the patient's death. We define the Markov kernel  $Q$  (case  $h = 3$  is omitted as no jumps are allowed when patients are dead) for all real-valued bounded measurable functions  $g$  on  $E$  :

$$\begin{aligned} Q_{(0,k)}^1 g(x) &= g(2, k, \zeta_0, 0, \tau), \\ Q_{(1,k)}^1 g(x) &= g(2, k, \zeta, 0, \tau) \mathbb{1}_{\zeta > \zeta_0} + g(0, k, \zeta_0, 0, \tau) \mathbb{1}_{\zeta = \zeta_0}, \\ Q_{(2,k)}^1 g(x) &= g(3, k, D, 0, \tau) \mathbb{1}_{\zeta = D}. \end{aligned}$$

Under both dynamics, one could write the transition kernel  $\mathcal{P}^\ell$  as a combination of the deterministic flow, the jump intensity and the Markov kernel such that for a time period  $r$  and all real-valued bounded measurable functions  $g$  on  $E$ ,  $\mathcal{P}^\ell g(x) = \mathbb{E}[g(X_r)|X_0 = x, \ell]$ .

The transition kernel  $\mathcal{P}$  can be written as a combination of the deterministic flow, the jump intensity and the Markov kernel such that for a time period  $r$  and all real-valued bounded measurable functions  $g$  on  $E$ ,  $\mathcal{P}g(x) = \mathbb{E}[g(X_r)|X_0 = x]$ . It is assumed that in the event of a relapse from the remission state ( $m = 0$ ), the patient takes more than 60 days to die ( $t_{(0,k)}^{0,*}(\mathbf{x}) > 60$  and  $t_{(2,k)}^{\ell,*}(\mathbf{x}) > 60$ ), this implies  $v_1 < 0.061$  and  $v_2 < 0.061$ . As this form of  $\mathcal{P}^\ell$  is complex, we work case by case for  $x = (\mathbf{m} = (h, k), \mathbf{x} = (\zeta, u, \tau, t)) \in E$ .

- If  $\mathbf{m} = (3)$ , then  $\mathcal{P}^\ell g(x) = g(3, k, D, 0, \tau)$ , the patient is dead.
- If  $\mathbf{m} = (2, k)$ , over a time step  $r$  the patient may either die or remain in therapeutic escape.

$$\mathcal{P}^\ell g(x) = g(3, k, D, 0, \tau) \mathbb{1}_{t_m^{\ell, \star}(\zeta) \leq r} + g(\mathbf{m}, \Phi_m^\ell(x, r)) \mathbb{1}_{t_m^{\ell, \star}(\zeta) > r}.$$

- If  $\mathbf{m} = (1, k)$ , and no treatment is applied ( $\ell = 0$ ), the biomarker increases. Over a time step  $r$ , the patient may die, remain in relapse, transition to a therapeutic escape and stay there, or transition to a therapeutic escape and die.

$$\begin{aligned} \mathcal{P}^0 g(x) &= g(3, k, D, 0, 0) e^{-\Lambda_m^\ell(x, t_m^{\ell, \star}(\zeta))} \mathbb{1}_{t_m^{\ell, \star}(\zeta) \leq r} + g(\mathbf{m}, \Phi_m^\ell(x, r)) e^{-\Lambda_m^\ell(x, r)} \mathbb{1}_{t_m^{\ell, \star}(\zeta) > r} \\ &\quad + \int_0^r g\left(2, k, \Phi_{(2,0,k)}^\ell(\Phi_m^\ell(\zeta, s), r-s), r-s, 0\right) \\ &\quad \times \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(\Phi_m^\ell(\zeta, s)) e^{-\Lambda_m^\ell(x, s)} \mathbb{1}_{t_{(2,k)}^{\ell, \star}(\Phi_m^\ell(\zeta, s)) > r-s} \mathbb{1}_{t_m^{\ell, \star}(\zeta) > s} ds \\ &\quad + \int_0^r g(3, k, D, 0, 0) \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(\Phi_m^\ell(\zeta, s)) e^{-\Lambda_m^\ell(x, s)} \mathbb{1}_{t_{(2,k)}^{\ell, \star}(\Phi_m^\ell(\zeta, s)) \leq r-s} ds \end{aligned}$$

- If  $\mathbf{m} = (1, k)$ , and treatment is applied ( $\ell = 1$ ), the biomarker decreases. Over a time step  $r$ , the patient may fall in remission and remain there, remain in relapse, transition to a therapeutic escape and remain there, fall into remission and transition to a therapeutic escape, transition to a therapeutic escape and die.

$$\begin{aligned} \mathcal{P}^1 g(x) &= g(0, k, \zeta_0, r - t_m^{\ell, \star}(\zeta), \tau + r) e^{-\Lambda_m^\ell(x, t_m^{\ell, \star}(\zeta))} \mathbb{1}_{t_m^{\ell, \star}(\zeta) \leq r} \\ &\quad + g(1, k, \Phi_m^\ell(\zeta, r), u + r, \tau + r) e^{-\Lambda_m^\ell(x, r)} \mathbb{1}_{t_m^{\ell, \star}(\zeta) > r} \\ &\quad + \int_0^r g\left(2, k, \Phi_{(2,k)}^\ell(\Phi_m^\ell(\zeta, s), r-s), r-s, \tau + r\right) \\ &\quad \times \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(\Phi_m^\ell(\zeta, s)) e^{-\Lambda_m^\ell(x, s)} \mathbb{1}_{t_m^{\ell, \star}(\zeta) > s} \mathbb{1}_{t_{(2,k)}^{\ell, \star}(\Phi_m^\ell(\zeta, s)) > r-s} ds \\ &\quad + \int_0^{r-t_m^{\ell, \star}(\zeta)} g(2, k, \Phi_{(2,k)}^\ell(\zeta_0, r-s-t_m^{\ell, \star}(\zeta)), r-s-t_m^{\ell, \star}(\zeta), \tau + r) \\ &\quad \times \mu_{(0,k) \rightarrow (2,k)}^\ell(s) e^{-\Lambda_m^\ell(x, t_m^{\ell, \star}(\zeta))} e^{-\Lambda_{(0,k)}^\ell(x, s)} \mathbb{1}_{t_m^{\ell, \star}(\zeta) \leq r} ds \\ &\quad + \int_0^r g(3, k, D, 0, \tau) \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(\Phi_m^\ell(\zeta, s)) e^{-\Lambda_m^\ell(x, s)} \mathbb{1}_{t_m^{\ell, \star}(\zeta) > s} \mathbb{1}_{t_{(2,k)}^{\ell, \star}(\Phi_m^\ell(\zeta, s)) \leq r-s} ds \end{aligned}$$

- If  $\mathbf{m} = (0, k)$ , and no treatment is applied ( $\ell = 0$ ), the biomarker remains at its nominal value. Over a time step  $r$ , the patient may stay in remission, transition to a relapse, transition to a therapeutic escape or transition to a

relapse before transitioning to a therapeutic escape.

$$\begin{aligned}
\mathcal{P}^0 g(x) = & g\left(\mathbf{m}, \Phi_{(0,k)}^\ell(\mathbf{x}, r)\right) e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, r)} \\
& + \int_0^r g\left(1, k, \Phi_{(1,k)}^\ell(\zeta_0, r-s), r-s, 0\right) \mu_{\mathbf{m} \rightarrow (1,k)}^\ell(u+s) \\
& \quad \times e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, s)} e^{-\Lambda_{(1,k)}^\ell(\Phi_{(1,k)}^\ell(\zeta_0, r-s), r-s)} ds \\
& + \int_0^r g\left(2, k, \Phi_{(2,k)}^\ell(\zeta_0, r-s), r-s, 0\right) \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(u+s) e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, s)} ds \\
& + \int_0^r \int_0^{r-s} g\left(2, k, \Phi_{(2,k)}^\ell\left(\Phi_{(1,k)}^\ell(\zeta_0, s'), r-s'-s\right), r-s'-s, 0\right) \\
& \quad \times \mu_{\mathbf{m} \rightarrow (1,k)}^\ell(u+s) e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, s)} \mu_{(1,k) \rightarrow (2,k)}^\ell\left(\Phi_{(1,k)}^\ell(\zeta_0, s')\right) e^{-\Lambda_{(1,k)}^\ell((\zeta_0, 0), s')} ds' ds
\end{aligned}$$

- If  $\mathbf{m} = (0, k)$ , and treatment is applied ( $\ell = 1$ ), the biomarker remains at its nominal value. Over a time step  $r$ , the patient may stay in remission or transition to a therapeutic escape.

$$\begin{aligned}
\mathcal{P}^1 g(x) = & g\left(\mathbf{m}, \Phi_{(\mathbf{m})}^\ell(\mathbf{x}, r)\right) e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, r)} \\
& + \int_0^r g\left(2, k, \Phi_{(2,k)}^\ell(\zeta_0, r-s), r-s, \tau+r\right) \mu_{\mathbf{m} \rightarrow (2,k)}^\ell(u+s) e^{-\Lambda_{\mathbf{m}}^\ell(\mathbf{x}, s)} ds
\end{aligned}$$

Although the PDMP transition kernel appears lengthy to write out, its structure allows for straightforward simulation.

## 2 Simulation-based reinforcement learning on solving BAPOMDPs

This section presents technical details used to tune and train deep RL algorithms on solving the BAPOMDP in our medical patient follow-up application. Next, we describe the tuning of hyperparameters performed for each selected algorithm and finally we analyse the results of the numerical experiments, focusing on the performance of the trained policies.

### 2.1 Algorithm description

Regarding the environment, Table 1 show the parameters of the PDMP (the Patient model) and BAPOMDP models used in the experiments. Both models were implemented using the Gymnasium environment [1] to ensure compatibility with the main RL libraries. We use in experiments the implementation provided in the RLlib framework [2]. Finally, RL algorithms optimize rewards rather than costs. Since our reward is defined as the negation of the cost  $r(x, a, x') = -c(x, a, x')$ , maximizing reward is equivalent to minimizing cost, ensuring our results remain consistent with the original objective.

Table 1: **PDMP and BAPOMDP parameter values used in experiments**, where  $v_1 \sim \text{Log-}\mathcal{N}(\mu, \sigma)$  and  $(\mu, \sigma)$  are unknown

| Name                                                    | Notation                                                         | Value                             |
|---------------------------------------------------------|------------------------------------------------------------------|-----------------------------------|
| <b>Boundaries</b>                                       |                                                                  |                                   |
| Biomarker nominal level                                 | $\zeta_0$                                                        | 1                                 |
| Biomarker death level                                   | $D$                                                              | 40                                |
| Horizon                                                 | $T$                                                              | 2400                              |
| <b>Flow</b>                                             |                                                                  |                                   |
| Slope parameter in relapse                              | $v_1$                                                            | $\text{Log-}\mathcal{N}(-3.5, 0)$ |
| Slope parameter in non curable relapse                  | $v_2$                                                            | $U[0.0001, 0.06]$                 |
| <b>Jump function</b>                                    |                                                                  |                                   |
| Scale parameter of $\mu_{(0,k) \rightarrow (1,k)}^\ell$ | $\alpha_{(0,k) \rightarrow (1,k)}^\ell$                          | $0.0001/k$                        |
| Shape parameter of $\mu_{(0,k) \rightarrow (1,k)}^\ell$ | $\beta_{(0,k) \rightarrow (1,k)}^k$                              | 3.5                               |
| Scale parameter of $\mu_{(0,k) \rightarrow (2,k)}^\ell$ | $\alpha_{(0,k) \rightarrow (2,k)}^k$                             | 0.000005                          |
| Shape parameter of $\mu_{(0,k) \rightarrow (2,k)}^\ell$ | $\beta_{(0,k) \rightarrow (2,k)}^k$                              | 3.5                               |
| Scale parameter of $\mu_{(1,k) \rightarrow (2,k)}^0$    | $\alpha_k^0$                                                     | $1.116e^{-20}k/4$                 |
| Shape parameter of $\mu_{(1,k) \rightarrow (2,k)}^0$    | $\beta_k^0$                                                      | 4.5                               |
| Scale parameter of $\mu_{(1,k) \rightarrow (2,k)}^1$    | $\alpha_k^1$                                                     | $5.58e^{-7}k$                     |
| Shape parameter of $\mu_{(1,k) \rightarrow (2,k)}^1$    | $\beta_k^1$                                                      | -0.8                              |
| <b>BAPOMDP</b>                                          |                                                                  |                                   |
| Noise                                                   | $\epsilon \sim \mathcal{N}(\mu, \sigma) \ (\mu = 0, \sigma = 1)$ |                                   |
| Non informative prior                                   | $\theta_0$                                                       | $(1, 0.001, 1.01, 1)$             |
| Weak prior                                              | $\theta_1$                                                       | $(-3.70, 6.001, 12.01, 1.15)$     |
| Medium prior                                            | $\theta_2$                                                       | $(-3.61, 12.001, 7.01, 1.15)$     |
| High prior                                              | $\theta_3$                                                       | $(-3.51, 36.001, 19.01, 1.80)$    |
| <b>Cost function</b>                                    |                                                                  |                                   |
| Treatment cost                                          | $C_\ell$                                                         | 0.1                               |
| Visit cost                                              | $C_v$                                                            | 1                                 |
| Death cost                                              | $C_D$                                                            | 330                               |
| Horizon constraint cost                                 | $C_H$                                                            | $10^6$                            |
| Treatment constraint cost                               | $C_T$                                                            | $10^6$                            |

## 2.2 Hyperparameter Tuning

We performed a hyperparameter tuning process using the Ray Tune functionality from the RLlib library to identify the best-performing configuration. The set of hyperparameters considered in this tuning is reported in Table 2. For each algorithm, 1000 configurations were randomly sampled from the ranges specified in Table 2 and trained in the training environment (the RL simulator of the BAPOMDP model). Training for each configuration was terminated once it reached either 100 000 simulation timesteps or 1000 training iterations, whichever occurred first. The best configuration for each algorithm was defined as the one achieving the highest average reward among all tested configurations. Table 3 summarizes the selected hyperparameters for each algorithm.

Table 2: Summary of tuned parameters and their possible values.

| Parameter                       | Possible Values                      |
|---------------------------------|--------------------------------------|
| Discount Factor ( $\gamma$ )    | {0.95, 0.97, 0.99, 0.999}            |
| Training Batch Size             | {32, 65, 256, 512, 1024, 2048, 4096} |
| Training Batch Mode             | {TruncateEpisodes, CompleteEpisodes} |
| Learning Rate                   | {3e-5, 0.0001, 0.0003, 0.001}        |
| Fully Connected Layer Sizes     | {[32], [32,32], [64], [64,64]}       |
| Activation Function             | {linear, relu, tanh}                 |
| Observation Filter              | {meanStdFilter, NoFilter}            |
| TD Error Loss                   | {Huber, MSE}                         |
| Number of Step                  | {1, 2, 3}                            |
| Double Q-learning               | {True, False}                        |
| Dueling Architecture            | {True, False}                        |
| Replay Buffer Capacity          | {50000, 100000, 500000, 1000000}     |
| Prioritized Replay Alpha        | {0.5, 0.6}                           |
| Prioritized Replay Beta         | {0.4, 0.5}                           |
| Prioritized Replay Epsilon      | {1e-6, 3e-6}                         |
| Epsilon Timestep                | {2, 10000, 50000, 100000, 200000}    |
| Initial Epsilon                 | {0.9, 1.0, 1.5}                      |
| Final Epsilon                   | {0.0, 0.01, 0.02}                    |
| Value Range Min                 | {-60 000, -50 000, -40 000}          |
| Value Range Max                 | {-500, -10, 0, 10}                   |
| Noisy Nets                      | {True, False}                        |
| Hidden Layers Size              | {[64], [128], [256], [512]}          |
| Training Intensity              | {1, 4, 16, 32}                       |
| Target Network Update Frequency | {500, 1000, 5000, 10000, 20000}      |
| Steps Before Learning Starts    | {1000, 10000, 20000}                 |

Table 3: Best configuration for each algorithm trained with BAPOMDP framework.

| Parameter                       | Without AM         |
|---------------------------------|--------------------|
| TD Error Loss                   | MSE                |
| Discount Factor ( $\gamma$ )    | 0.99               |
| Number of Step                  | 1                  |
| Double Q-learning               | False              |
| Dueling Architecture            | False              |
| Replay Buffer Capacity          | 100 000            |
| Prioritized Replay Alpha        | 0.6                |
| Prioritized Replay Beta         | 0.4                |
| Prioritized Replay Epsilon      | $1 \times 10^{-6}$ |
| Epsilon Timestep                | 10000              |
| Initial Epsilon                 | 1.5                |
| Final Epsilon                   | 0.0                |
| Value Range Min                 | -40 000            |
| Value Range Max                 | -500               |
| Training Batch Size             | 256                |
| Training Batch Mode             | TruncatedEpisodes  |
| Learning Rate                   | 0.0003             |
| Fully Connected Layer Sizes     | [64,64]            |
| Activation Function             | linear             |
| Observation Filter              | NoFilter           |
| Noisy Nets                      | True               |
| Hidden Layers Size              | [128]              |
| Training Intensity              | 32                 |
| Target Network Update Frequency | 500                |
| Steps Before Learning Starts    | 10 000             |



### 2.3 Evaluation

For the evaluation, each algorithm was trained using the best hyperparameter configuration identified in the previous section. To account for randomness, five independent training runs were performed per algorithm. Training was terminated once it reached either 100,000 simulation timesteps or 1,000 training iterations, whichever occurred first. During training, the current policy was evaluated every five iterations in the environment without exploration noise. Each evaluation consisted of 10 episodes, and the performance metric was the average reward per episode under the current policy.

We report the performance of algorithm, using its best configuration, by plotting the mean episode reward as a function of the number of training interactions. In Figure

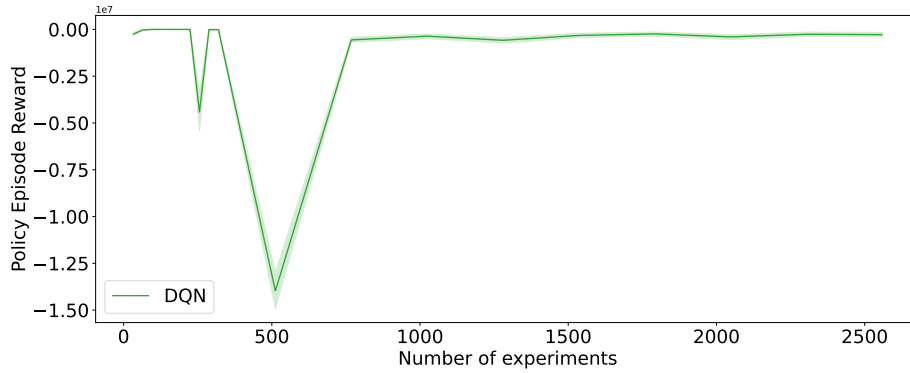


Fig. 1: Evaluation policies mean reward according to training experiences for DQN algorithm.

### References

1. Towers, M., Kwiatkowski, A., Terry, J., Balis, J.U., De Cola, G., Deleu, T., Goulão, M., Kallinteris, A., Krimmel, M., KG, A., et al.: Gymnasium: A standard interface for reinforcement learning environments. arXiv preprint arXiv:2407.17032 (2024)
2. Wu, Z., Liang, E., Luo, M., Mika, S., Gonzalez, J.E., Stoica, I.: RLlib flow: Distributed reinforcement learning is a dataflow problem. In: Conference on Neural Information Processing Systems (NeurIPS) (2021)